Exercise 8

EEPROM basics (Microchip Studio) (UNO)

In this task, you are going to read and write data to the UNOs EEPROM using two buttons and the USART.

The system writes different data to the EEPROM. The data is selected from two char arrays. The selected array changes every time the write button is pressed.

Use a 32-byte array to store the user data that is being written. This written data is read when the button (pin 8) is pressed.

Use the USART of the previous exercise 1 to print out the EEPROM data via a terminal, such as **PuTTY**.

You can test that the data is stored in EEPROM by resetting or disconnecting the board for a while, and then pressing the buttons to read/write the data

Build the circuit using these components:

- Arduino UNO
- 2 x buttons
- 2 x 10 kΩ resistors
- Wires

First button is used to READ the data:
Connect one leg to ground via a 10 kΩ resistor, and the same leg is also connected to pin 8 of Arduino UNO, while the other leg is connected to power.

Second button is used to WRITE the data:
Connect one leg to ground via a 10 kΩ resistor, and the same leg is also connected to pin 5 of Arduino UNO, while the other leg is connected to power.

<u>Code in Microchip Studio:</u>

1. Build communication between your computer and Arduino UNO using USART. The step is similar to the Exercise 7.
2. Additionally, `#include <stdbool.h>` because Boolean functions are used. Also, `#include <string.h>` because the `strcpy` function is used.
3. **In the <u>main(void):</u>**
   - After you initialize the UART and redirect the `stdin` and `stdout`, set the pin and port for those two buttons:

     ```
     DDRB &= ~(1 << PB0); // pin 8 for read
     DDRD &= ~(1 << PD5); // pin 5 for write
     ```

   - Create the variable for those two buttons:

     ```
     bool button_read = 0;
     bool button_write = 0;
     ```

   - Use a 32-byte array to store the user data is being written:

     ```
     uint16_t max_address = 32;
     ```

- Setup the arrays for reading and writing in the EEPROM:
```
char input_data[32]; // used when reading the data from EEPROM
char output_data[32] = "Initial sentence"; // data to be written to EEPROM. Write your own sentence here
```

- Setup the data_1 and data_2 that will be written in the EEPROM:
```
char output_data_1[32] = "First sentence"; // Write your own sentence here
char output_data_2[32] = "Second sentence"; // Write your own sentence here
```

- Setup the switching between the data_1 and data_2:
```
bool b_switch_text = 0;
```

4. In the <u>while loop:</u>

- Setting read and write button:
```
button_read = (PINB & (1 << PB0));
button_write = (PIND & (1 << PD5));
```

- Create the code for reading the EEPROM when the **button_read** is pressed. Use EECR, EEAR, and EEDR register. Read the datasheet p. 22 – 24 and example in p. 26.
```
if (1 == button_read)
{
      for (uint16_t address_index = 0; address_index < max_address; address_index++)
      {

          while(EECR & (1 << EEPE))//datasheet p.22 and see example in datasheet p. 25
          {
          /* wait for the previous write operation to end */
          }

          EEAR  = address_index;//datasheet p.26
          EECR  |= (1<<EERE); //datasheet p.22 and see example in datasheet p. 26
          input_data[address_index] = EEDR; //read datasheet p.22. EEDR contains the data
          READ OUT from EEPROM.
          }
```

- Print out the Read data
```
          printf(input_data);
}
```

- Create the code for writing the EEPROM when the **button_write** is pressed. Use EECR, EEAR, and EEDR register. Read the datasheet p. 22-24 and example in p.25.
```
else if (1 == button_write)
{
      for (uint16_t address_index = 0; address_index < sizeof(output_data); address_index++)
      {
          while(EECR & (1 << EEPE))//datasheet p.22 and see example in datasheet p. 25
          {
          /* wait for the previous write operation to end */
          }

          EEAR = address_index;//datasheet p.22
          EEDR = output_data[address_index] ;//EEPROM data register, datasheet p.22. In write,
          EEDR contains data TO BE WRITTEN to EEPROM.
          EECR |= (1 << EEMPE); // master write enable, datasheet p.22 and p.23
          EECR |= (1 << EEPE); // enable EEPROM write, datasheet p.22 and p.23
          }
```

- Print out the Write data
```
          printf(output_data);
}
```

- Create the code to switch between data_1 and data_2 when **button_write** is pressed. Use *stcrpy* function to copy data from string into the memory of another string.

```
if(0 == b_switch_text)
{
    strcpy(output_data, output_data_2); //function copies data from one string into the memory
    of another string.
    b_switch_text = 1;
}
else if (1 == b_switch_text)
{
    strcpy(output_data, output_data_1);
    b_switch_text = 0;
}
```

NOTE: After you build the code, upload and print out the data using PuTTY. Press the write button and read button to see the result on the screen.